

# JAY KUBO

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Applied AI architect in Tokyo. A decade advising Japanese enterprises from inside Google Cloud, CrowdStrike, and Rakuten—now building agent orchestration and LLM evaluation on a 20-node cluster across four sites. Native Japanese and English, technical register.

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## EXPERIENCE

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### Founder, kub0 LLC

*Tokyo, Japan | Jul 2020 - Present*

#### AGENT ORCHESTRATION

- Claude Code distills work to Grok Build workers under kernel sandboxes; a write needs a YubiKey grant. Done is a re-run predicate, not the worker's self-report.
- The same hard gates ship in the executor image as a deterministic PreToolUse hook: warn mode first, then flipped to enforce on the executor pods; the orchestrator's own dispatch-time check is the next stage.
- Claude Code TUI status line for remaining context (advisory); same on Grok sessions. Gaius as a Claude Code plugin (OSS, PyPI, MCP). ~30 skills, including a scoped-diff security audit that stays inside the context window.
- Project Gaius: 10,000+ facts from Claude and Gemini sessions. 98.6% retrieval hit@10 on LongMemEval-S, offline, no GPU.

#### EVALUATION

- LLM-as-judge (base, RAG, fine-tuned) across two model lineages. Fine-tune scored 2.79 vs RAG 3.92, so I shipped RAG. Judge agreement measured 47% against a 70% bar, so I shipped on the margin, not the grade.
- Found alerts that passed review and could never fire (always-zero compare; missing credential). Adversarial harness: 85 detected, 5 bypassed, all test debt.

#### PLATFORM

- Packaged Kubernetes security product for GCP Marketplace (Helm, GKE). Approved in Anthropic's Cyber Verification Program.
- Daily kube-bench (k3s-cis-1.9) into a CIS posture API: living hardening state, not a one-time audit.
- Cross-compiled a custom ARM64 kernel with BPF\_LSM, absent from the stock kernel, to run eBPF enforcement on ARM nodes.
- 20-node Kubernetes cluster, Austin, Los Angeles, Torrance, Tokyo. Ansible, WireGuard, ~1 TB GPU memory, 47 TB DRBD-replicated block (NVMe + SATA). OpenTelemetry/Grafana, two-proxy OAuth2. Site outage shifts traffic automatically.
- vLLM serving Gemma, Qwen, and an embedding model on AMD ROCm and NVIDIA CUDA. LoRA fine-tuning on a DGX Spark.

### Technical Advisor, Japan Deluxe Tours, Inc.

*Remote | May 2015 - Present*

- Claude in production for their staff: a Google Chat bot that answers operations questions and ships approved changes, gated by Google Groups roles (admin vs staff) and logged. The same roles gate ChatOps merges and experimental features.
- Recommended and stood up their first AWS cloud in 2015. Cut RDS cost ~50% while still on AWS. Migrated the customer site to Cloudflare Workers and on-prem Kubernetes.
- MySQL 8.0 on NVMe with GTID replication, two streaming replicas at 0 s lag.
- Stripe payments (deposits and self-service balance, no stored card). Sign in with Apple and Google on the customer site.
- Legacy staff tooling behind Cloudflare Access. Staff Macs on Tailscale; FleetDM for inventory and CVE.

### Associate Analyst, CrowdStrike (Falcon Complete Japan)

*Austin, TX | Jan 2025 - May 2025*

- Japanese-language translator on high-priority Falcon Complete Japan escalations—analysts and senior analysts pulled me in, and what I wrote is what went out. Also worked with a GovCloud senior analyst on Japan commercial-cloud operations.
- Claude workflow for Japanese customer responses (SLA docs, JIRA, Confluence). Daily threat-intel feeds and ~30 SOC automations the team kept.

### Application Support Analyst, EPAM Systems (Google Cloud)

*Austin, TX | Nov 2019 - Mar 2021*

- Frontline support for Japanese enterprise customers on Google Cloud, customer-facing under Google: BigQuery, Dataflow, Cloud Composer, Cloud Spanner, Cloud SQL, App Engine, Dialogflow. Gaming, e-commerce, rail, trading houses, telecom, retail, banking.
- Bridged U.S. and Japan on high-priority escalations. Chrome extension cut Platinum P0/P1 SLA misses to near zero.
- Carried Won't Fix rulings back to enterprise customers; moved stalled private cases to the public tracker.

- Filed P2s in Google Issue Tracker from those accounts, including: Cloud SQL export blocked for viewer IAM, Dataproc fluentd leak on the master until OOM, billing accounts missing in Console, Cloud SDK output drift between Python 2 and 3.

## Security Engineer & Data Analyst, Rakuten

*San Mateo, CA | Sep 2015 - Jul 2018*

- Created an internal vulnerability-intelligence platform to collect, search, alert, and visualize CVEs in real time. Sold across subsidiaries (Ebates, Slice, Buy.com, Linkshare). Whitebox and blackbox pentests for the same brands.
- Kafka pipeline: Palo Alto PAN-OS syslog into Elasticsearch. Authored the company MySQL and Oracle security standards from CIS benchmarks, across thousands of instances. On-call DBA for 100+ Japan-region services (~5,000 VMs).
- Azure pipelines and DOMO dashboards. A/B and multivariate tests in Oracle Maxymiser across six markets. 4-month OJT with the Rakuten HQ data-science team in Tokyo (CNN product-image matching, OpenCV, Chainer).
- Led Rakuten's first PII/SPII sweep for GDPR compliance, embedded with the DBA team as a dual post across the company's full database landscape. Coordinated with thousands of internal teams, much of it in Japanese.

## EDUCATION

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B.A., Mathematics and Computer Science (Courant Institute) · New York University · May 2015

Coursework: Artificial Intelligence · Algorithmic Problem Solving · Database Systems (Courant, graduate) · Mathematics of Finance · Meaning of Leadership (Wagner) · Digital Marketing and Social Media Analytics (Stern)

## CERTIFICATIONS

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Certified Falcon Administrator (CCFA) · CrowdStrike · Apr 2025 - Apr 2028

Remote Pilot Certification (Part 107) · FAA · Issued Oct 2025

Advanced Open Water Diver · PADI · Issued Aug 2025

Private Pilot License · FAA · Issued Dec 2024

Continuous Monitoring Certification (GMON) · GIAC · Expired May 2026

## TRAINING

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Assessing and Exploiting Control Systems and IIoT · Black Hat · Apr 2024

Abusing and Protecting Kubernetes, Linux and Containers · Black Hat · Mar 2024

Reverse Engineering Firmware with Ghidra · Black Hat · Dec 2023

Machine Learning (inaugural) · Black Hat · Aug 2023

Reversing Signal with Software-Defined Radio · Black Hat · Aug 2023

Advanced Infrastructure Hacking · Black Hat · Mar 2023

ICS/SCADA Security Essentials · SANS · Dec 2022

A Crash Course in Practical Fast Forensics with a Red Teaming Perspective (Japanese) · Black Hat · Nov 2022

## OPEN SOURCE & WRITING

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PyPI gaius-memory · Ubuntu kernel bug LP#2150798 · cluster engineering writing at jkubo.com

## SKILLS

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**Spoken languages:** Japanese (native; business and technical) · English (native; business and technical)

**AI:** Claude API · Claude Code · Grok Build · Gemini · RAG · LLM-as-judge eval · vLLM · LoRA · MCP · Python · Go

**Cloud:** GCP (BigQuery, Dataflow, Cloud Composer, Cloud Spanner, Cloud SQL, GKE, Marketplace, Dialogflow) · AWS · Azure · Kubernetes · Ansible · OpenTelemetry/Grafana

**Security:** Linux · eBPF · YubiKey · endpoint · threat intelligence · penetration testing